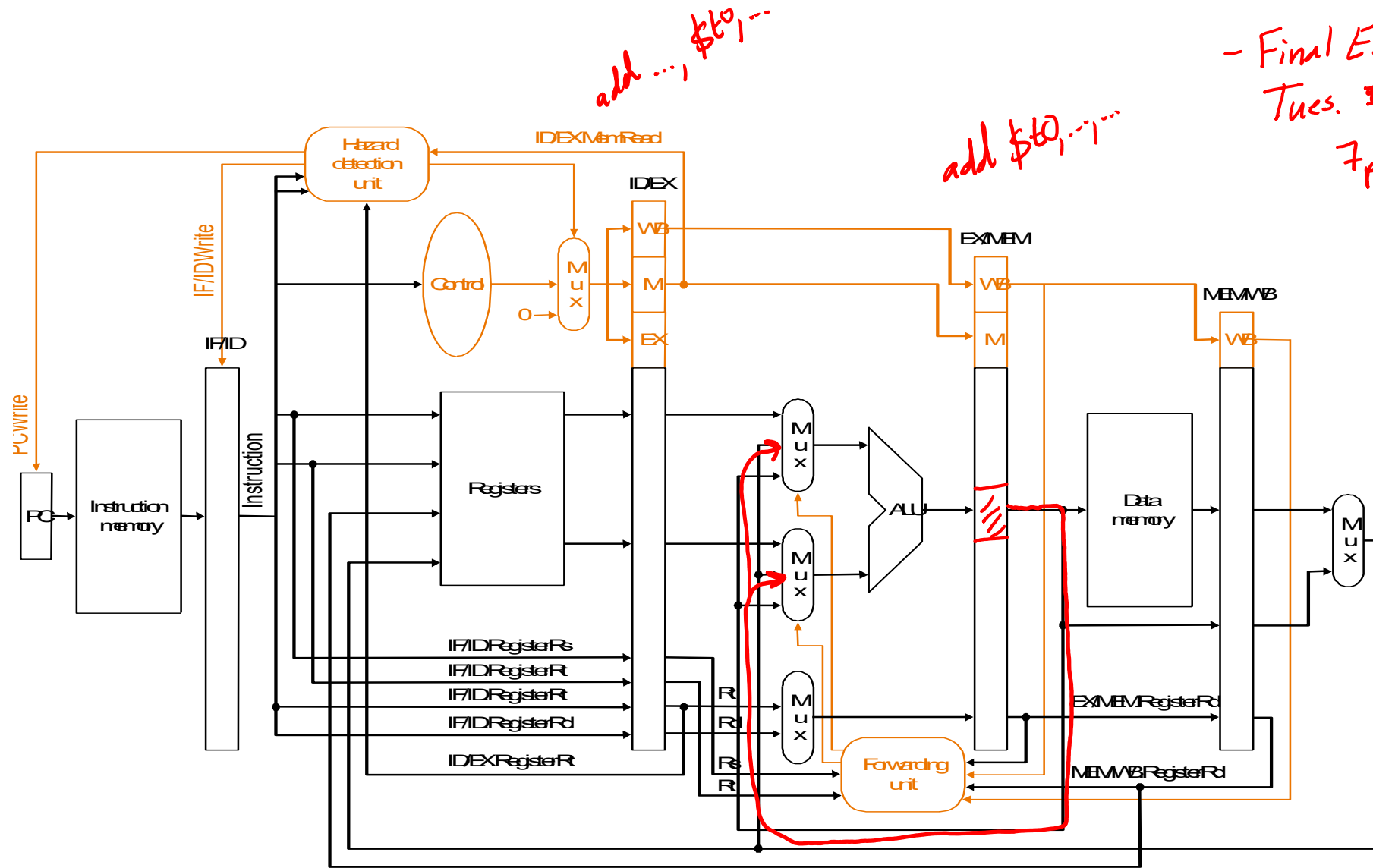




- Final Exam
Tues. ~~4p~~ (14-303)
7p-8p

Example 1

A program is composed of 25% loads, 25% stores, 30% ALU inst, and 20% branches. Assume 30% of the loads are followed by an instruction that uses the load result. Assume branches are resolved in the **ID** stage and 40% of the branches are taken (next instruction is always fetched). What is the average CPI for the program?

$$\boxed{1.155 \text{ CPI}} = \overbrace{.25 (.7 (1) + .3 (1+1))}^{\text{lw}} + \overbrace{.25 (1)}^{\text{sw}} + \overbrace{.30 (1)}^{\text{ALU}} + \overbrace{.20 (.40 (1+1) + .60 (1))}^{\text{branches}}$$

Handwritten annotations for the branches term:
- "1 squash" with an arrow pointing to the $(1+1)$ term.
- "taken" with an arrow pointing to the $.40 (1+1)$ term.
- "not taken" with an arrow pointing to the $.60 (1)$ term.

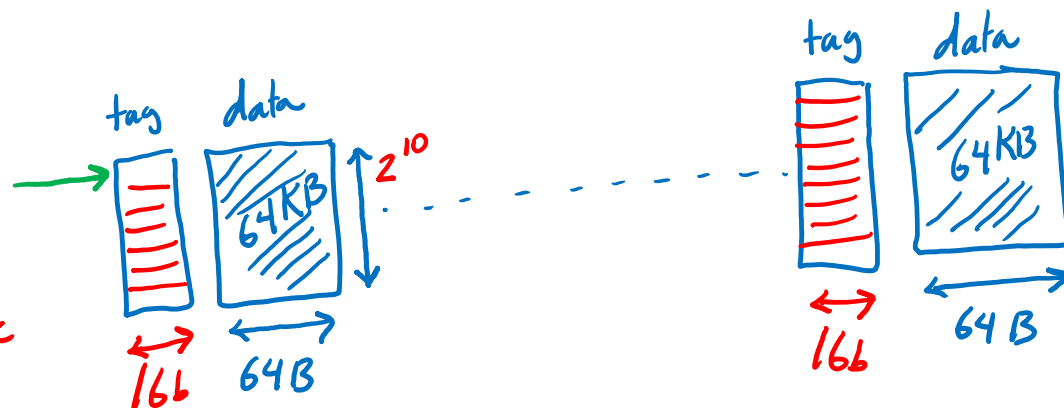
NS2

- L3 = 4 MB
- 8-way
- 64 B block
- 32 b addr
- byte addr

Nintendo Switch L2

- 1 MB = 1024 KB
- 16-way
- 64 B blocks
- 32-bit addresses

- byte addressable



1. Block offset = $64B \rightarrow 16W \rightarrow 2^4W \rightarrow 4 \text{ bits}$

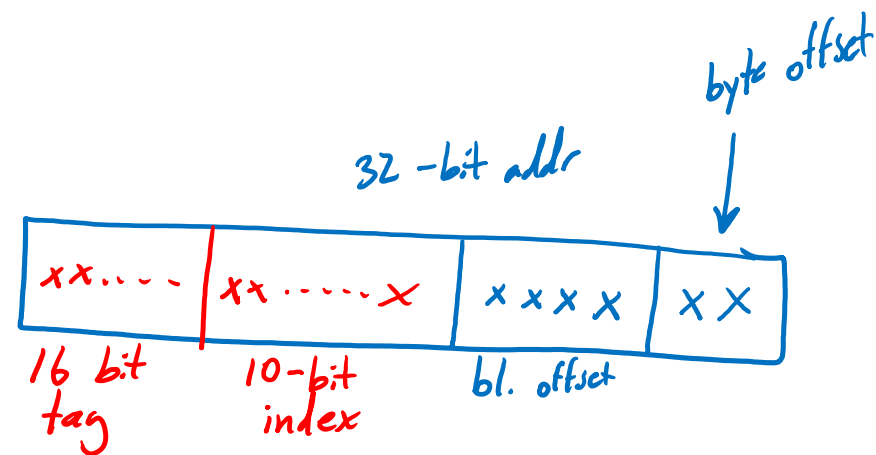
2. Index : $\frac{1024 \text{ KB}}{16} = \frac{2^{10} \text{ KB}}{2^4} = 64 \text{ KB}$

$\frac{64 \text{ KB}}{64 \text{ B}} = \frac{2^6 \cdot 2^{10} \text{ B}}{2^6 \text{ B}} = 2^{10} \text{ indices} \rightarrow 10 \text{ bit index}$

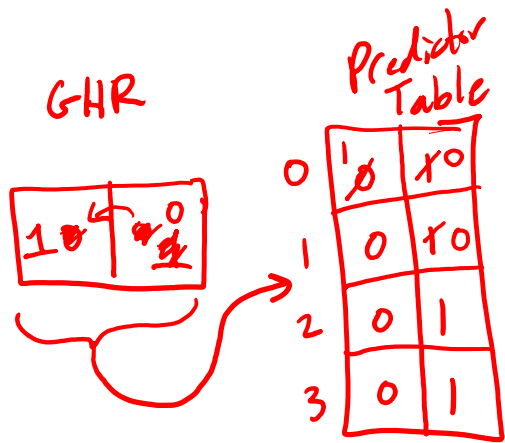
3. Tag : $32 - 10 - 4 - 2 = 16 \text{ bit}$

4. Total tag size :

$16 \text{ bits} \times 2^{10} \times 16 = 2^4 \text{ b} \times 2^{10} \times 2^4$
 $= 2^{18} \text{ bits}$



init table to 01



prediction	actual	correct?
NT	T	X
NT	NT	✓

Things to know for the final (after midterm)

- Chapter 4.5-4.8
 - Understand pipelined datapath
 - Compute CPI for pipelined processors
 - Branch hazards, squashing, and lab 4 material
 - Data hazards (EX and MEM hazards)
 - Forwarding and stalls
 - Branch predictors

- You will not need to draw anything on the final

Things to know for the final (after midterm)

- Chapter 5.1-5.5
 - How to find data (byte offset, block offset, index, tag)
 - How to compute hits and misses
 - direct ^(1-way) mapped, associativity, LRU, write-through, write-back, valid bit, dirty bit
- Any terms in the book, but not covered in class will not be on the exam